RAJALAKSHMI ENGINEERING COLLEGE

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A2

Started on	Tuesday, 29 September 2026, 8:59 AM
State	Finished
Completed on	Thursday, 8 October 2026, 9:06 AM
Time taken	9 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2		**g**	x	**t**	**x**	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s1[1000], s2[1000];

    scanf("%s", s1);
    scanf("%s", s2);

    int n = strlen(s1);
    int m = strlen(s2);

    int dp[n + 1][m + 1];

    // Initialize first row and column
    for (int i = 0; i <= n; i++) {
        dp[i][0] = 0;
    }

    for (int j = 0; j <= m; j++) {
        dp[0][j] = 0;
    }

    // Build the DP table
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = (dp[i - 1][j] > dp[i][j - 1])
                           ? dp[i - 1][j]
                           : dp[i][j - 1];
            }
        }
    }
```

```
36
37      printf("%d\n", dp[n][m]);
38
39      return 0;
40  }
41
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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